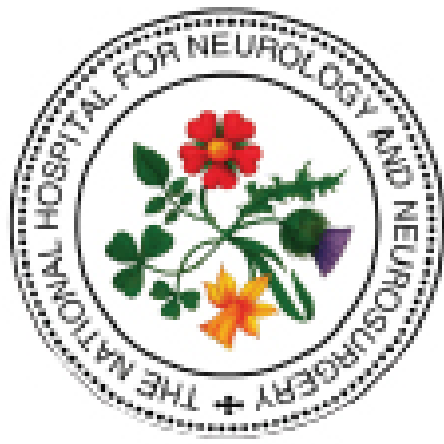




2. DIAGNOSING DEATH by NEUROLOGICAL CRITERIA (DNC) – Preconditions / Exclusions

Academy of Medical Royal Colleges (2025) Code of Practice in UK

PRECONDITIONS

Specific preconditions for the application of neurological criteria must be fulfilled before doctors commence clinical testing

- Aetiology severe enough to cause permanent cessation of brainstem function
Should be no doubt that condition due to devastating brain injury of known aetiology / mechanism
- Assessment period sufficient to exclude potential for recovery
*In primary intracranial pathology at least 6h following loss of last observed brainstem reflex or spontaneous breath
In acute hypoxic-ischaemic encephalopathy or post cardiorespiratory arrest at least 24h following loss of last observed brainstem reflex or spontaneous breath*
- Exclusion of potentially reversible factors contributing to coma or apnoea
Factors which risk confounding diagnosis of death by neurological criteria must be excluded (see Exclusions below)
- Caution diagnosing DNC in uncommon circumstances
 - If aetiology primarily posterior fossa / brainstem then MRI recommended prior to DNC
 - Initially isolated posterior fossa lesion may develop supratentorial pathology and standard approaches to neurological criteria would then apply
 - Agreement in MDT that no further appropriate therapeutic options of benefit and enough time has elapsed to exclude possibility of neurological recovery

EXCLUSIONS

Hypothermia

- Core temperature $\geq 36^{\circ}$ C at time of clinical testing - may require active warming
 - Hypothermia may delay return of brainstem function - neurological criteria should not be applied until at least 24h following correction of hypothermia (core temperature of $\geq 36^{\circ}$ C)
 - Following correction of hypothermia any transient and temporary reductions in temperature to below 36° C do not mandate further 24h observation at normothermia

Depressant drugs

- Patients with devastating brain injury are considered as having a vulnerable brain which might be more sensitive to lower levels of depressant drugs
- Exclude possibility of ongoing drug effect being cause of, or materially contributing to, coma or apnoea

Profound neuromuscular weakness

- Neuromuscular weakness resembling absence of brainstem reflexes may occur as consequence of neuromuscular or metabolic disorders, including critical illness neuromyopathies, or be the result of the effect of some drugs:
 - Neuromuscular blocking agents and other drugs must be excluded
 - Peripheral nerve stimulator or presence of deep-tendon reflexes, electromyography to confirm that adequate neuromuscular function present
 - Where pre-existing or acquired neuromuscular disorder known or possible, consideration given to impact of drug history and muscle function before applying neurological criteria

Cervical spinal cord pathology

- If **high** cervical cord pathology +/- associated cervical spine injury, contributing to apnoea, further investigation (CT or MRI) needed
 - Where imaging demonstrates cervical spinal cord pathology, apnoea test alone cannot reliably confirm loss of capacity to breathe and must be supplemented with an ancillary investigation
- In cervical spinal cord pathology, apnoea test can still be performed unless injury of such an extent that spontaneous breathing considered impossible

Circulatory and respiratory disturbances

- Normal (or patient baseline) physiological parameters prior to and during testing
- Doctors undertaking clinical testing satisfied that any disturbance is not materially contributing to the observed coma or apnoea

EXCLUSIONS

Electrolyte and metabolic disturbances

METABOLIC / ENDOCRINE	TARGET
HYPOTENSION	MAP >60 mmHg
HYPOTHERMIA	Minimum Core Temperature 36° C
HYPO / HYPERNATRAEMIA	Na ⁺ 125-160mmol/L (inclusive)
HYPOKALAEMIA	K ⁺ >2 mmol/L
HYPO / HYPERPHOSPHALAEMIA	PO ₄ 0.5-3.0mmol/L (inclusive)
HYPO / HYPERMAGNESAEMIA	Mg ²⁺ 0.5-3.0mmol/L (inclusive)
HYPO / HYPERGLYCAEMIA	BG 3--20mmol/L (inclusive)
ACIDAEMIA / ALKALAEMIA	pH 7.35-7.45 / H ⁺ 45-35mmol/L

Sodium

- Effects of hyponatraemia depend on rate of development, but rare for patients to become unresponsive if sodium >115 mmol/L
 - If severe hyponatraemia corrected too rapidly patient may develop osmotic demyelination syndrome, which can cause potentially reversible coma
 - Sodium levels above 160 mmol/L can be associated with unresponsiveness

Potassium

- Very low levels of potassium may cause myopathy and levels <1.0 mmol/L have been reported to cause flaccid quadriplegia

Phosphate and magnesium

- Profound elevation or lowering of phosphate or magnesium may be associated with severe neuromuscular weakness that may culminate in flaccid quadriplegia

Glucose

- As blood glucose concentrations can change rapidly in the critically ill, blood sugar measurement should be made prior to clinical testing
 - Hyperglycaemia in diabetic ketoacidosis or hyperosmolar hyperglycaemic state may cause a state of unresponsiveness and cessation of brainstem function, but extremely unlikely with blood glucose levels ≤ 20.0 mmol/L
 - Severe hypoglycaemia is associated with coma or stupor and testing of brainstem reflexes should not be undertaken if glucose level below 3.0mmol/L

Other electrolyte and metabolic disturbances

Disturbances may rarely contribute to coma or apnoea

- Disturbances such as elevated urea or ammonia, metabolic disorders
- Consider correction, expert metabolic advice, or ancillary investigation

Endocrine disturbances

Should be satisfied no clinical reason to suspect an endocrine disturbance is materially contributing to coma or apnoea - If doubt, endocrine assays undertaken

- Myxoedema may cause a deep unresponsive coma
- Patients in thyroid storm may present in acute coma or with acute thyrotoxic myopathy
- Addisonian crisis may be associated with severe neuromuscular weakness causing an acute ascending paralysis or encephalopathy proceeding to coma

‘RED FLAG’ GROUPS

Diagnostic caution advised in the following

‘RED FLAG’ groups

Consider need to delay testing and / or perform ancillary investigations

1. Testing <6 hours of loss of last brainstem reflex
2. Testing <24 hours of loss of last brainstem reflex, where aetiology primarily anoxic damage
3. If prior treatment has included hypothermia – 24 hours observation following re-warming to normothermia (36° C) recommended
4. Patients with any neuromuscular disorder
5. When steroids given in space occupying lesions, such as abscesses
6. Prolonged fentanyl infusions
7. Aetiology primarily located to brainstem or posterior fossa
8. Therapeutic decompressive craniectomy – with resultant change in intracranial dynamics
9. Associated upper cervical spinal cord injury

The Role of Ancillary Investigations

In certain situations, it may not be possible to satisfy all the preconditions or perform or complete all the clinical tests necessary to diagnose death using neurological criteria.

In such circumstances, ancillary investigation, either before or after clinical testing, is necessary to support a diagnosis of death using neurological criteria. No single ancillary investigation has worldwide consensus, though there has been a shift toward ancillary investigations that confirm the absence of cerebral blood flow or cerebral perfusion. The 2025 Code provides the principles for the use of ancillary investigations rather than to advise on the specifics of any individual ancillary investigation or protocol. Electroencephalography (EEG) is no longer recommended as an ancillary investigation for neurological criteria in the UK for adults or children.

UK consensus guideline for the use of cerebral CT angiography as an ancillary investigation to support a clinical diagnosis of death using neurological criteria in adults has been published.

The 2025 Code recommends that ancillary investigation is required in the following circumstances:

- When a comprehensive neurological examination, including the apnoea test, is not possible (e.g. high cervical cord pathology, inability to examine both eyes or both ears)
- When continuing effects of confounding factors which affect the preconditions cannot be excluded (e.g. residual sedation, metabolic or pharmacological derangement, decompressive craniectomy)

Ancillary investigation should also be considered when there is uncertainty regarding the interpretation of presumed non-brain mediated movements.